Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/29/2009
PI Name:	Trappe, Scott Ph.D.		
Project Title:	Influence of Exercise on Human Single Muscle Fiber Function with Long-Term Bedrest		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-05
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed back to 8/30/2007 per info from J. Dardano/JSC (6/08) NOTE: End date changed to 4/30/2009 per info from P. Denkins/JSC (8/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tesch, Per (Karolinska Institute) Trappe, Todd (University of Arkansas for Medical Sciences)		
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Task Description:	The purpose of this investigation will be to extend our previous spaceflight and ground-based studies that have examined the cellular mechanisms related to the loss of muscle mass and function with long duration unloading. More specifically, we will measure the contractile properties of individual slow- and fast-twitch muscle fibers from lower limb skeletal muscles and the concentration of the two main contractile proteins, myosin and actin, from single muscle fibers. These cellular physiology and molecular muscle studies will be conducted on subjects before and after 90-d of bedrest with and without countermeasure activities (aerobic and resistance training). These studies will provide further insight into the mechanisms of muscle wasting associated with spaceflight and provide information on effective countermeasures to maintain skeletal muscle mass and function in male and female space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[NOTE: Added by Task Book editor from final report; 6/2009] OVERALL STUDY DESIGN: This investigation integrated several physiologic, metabolic, and psychological experiments to examine the responses to prolonged bed rest (i.e. simulated weightlessness). RESULTS/DISCUSSION All data including tables and figures can be found in the articles we have published. We present here the take home messages from this research and a brief discussion on the two leg muscles that were investigated. TAKE HOME MESSAGE FOR HUMAN SPACE EXPLORATON: SKELETAL MUSCLE 1. Upper Leg – Complete and viable exercise countermeasures for whole muscle and single muscle fiber contractile function/structure. The resistance + aerobic exercise program was complimentary for unloaded skeletal muscle. 2. Lower Leg – Incomplete exercise prescription. The resistance + aerobic exercise program was better than resistance exercise alone. 3. Nutrition – The nutrition countermeasure provided no benefit for the protection of skeletal muscle during long-term unloading. In fact, the nutrition intervention led to greater loss in thigh muscle mass as compared to control. 4. Potential gender-related issues for skeletal muscle. Women appear to loss more muscle mass in a shorter period of time compared to men. 5. The WISE 2005 exercise prescription protocol and hardware were as good or better than what we have reported with the ISS regimen. 6. The last decade of exercise countermeasure research has significantly reduced the time necessary to protect most muscle groups (<2% total time in the unloaded state) for crewmembers. VASTUS LATERALIS This study shows that the concurrent training program of resistance (2-3 d/wk) and aerobic (3-4 d/wk) exercise was protective of the thigh muscle during 60-d of bed rest in women. This finding is supported by the maintenance of whole muscle and single muscle fiber parameters (MHC I and IIa) during the bed rest period. These data indicate that in light of the difficulty in protecting slow-twitch muscle fibers by resistance training alone during prolonged unloading, concurrent resistance and aerobic training protocols should be considered as a viable countermeasure for the upper leg of crewmembers during long duration space travel. The nutrition countermeasure did not provide an ergogentic benefit for whole muscle or single muscle fiber size and function during bed rest. This information should be useful to the international space community and health care industry working with patients stricken with extended bed rest. SOLEUS While the addition of aerobic exercise to resistance exercise proved complimentary and completely protective for the vastus lateralis, whole muscle and single muscle fiber properties for the soleus, in particular the MHC I fibers, did not achieve the same level of protection. The fact that MHC I fibers from the vastus lateralis and soleus muscle atrophy to a similar extent with long duration unloading, but respond differently to the same exercise loading program, was an interesting and novel finding. These data suggest that unique and specific exercise regimens for the upper (vastus lateralis) and lower leg (soleus) may be necessary to protect human skeletal muscle during extended unloading periods. Our impression is that the difference in the vastus lateralis and soleus muscle response to the concurrent exercise program is not a volume of exercise issue per se, but more related to the type of muscle contractions performed. All of the muscle contractions performed as part of this investigation (>100,000 per muscle group over 60-d) were dynamic in nature. However, given that the soleus is suited for postural support, static type contractions of longer duration to more closely mimic the typical load on Earth in a 1-g environment may be required. We do not recommend that static load contractions be performed in lieu of dynamic contractions. Rather, it is likely that a proper combination of static and dynamic contractions will be necessary to protect the quantitative (size) and qualitative (intrinsic speed and power) properties of the soleus muscle fibers.
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